Thomas Nguyen
11888002
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Problem 1. Data Distribution
Table 1 shows 2, 400 student grades and their preferences for courses collected from 10 years of university data.

	A	B	C	D	F
CPTS 412	341	504	383	117	25
CPTS 512	243	400	261	94	32

Table 1: Student grades and their preference on courses

a) Let p = $[p_A\ p_B\ p_C\ p_D\ p_F]^T$ be the probability distribution (frequency) of the course grade. find P for CS 412 and CS 512

CS 412:

$$341 + 504 + 383 + 117 + 25 = 1370$$

$$P^{(412)} = \begin{aligned} \tfrac{341}{1370} &= 0.249 \\ \tfrac{504}{1370} &= 0.368 \\ \tfrac{383}{1370} &= 0.280 \\ \tfrac{117}{1370} &= 0.085 \\ \tfrac{25}{1370} &= 0.018 \end{aligned}$$

CS 512:

$$243 + 400 + 261 + 94 + 32 = 1030$$

$$P^{(512)} = \begin{aligned} \tfrac{243}{1030} &= 0.236 \\ \tfrac{400}{1030} &= 0.388 \\ \tfrac{261}{1030} &= 0.253 \\ \tfrac{94}{1030} &= 0.091 \\ \tfrac{32}{1030} &= 0.031 \end{aligned}$$

b) Calculate the correlation value x^2 for "CS412" and "CS 512"

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Row totals:

$$\text{CS412} = 1370, \qquad \text{CS512} = 1030$$

Column totals:

$$A = 584, \quad B = 904, \quad C = 644, \quad D = 211, \quad F = 57$$

Grand total:

$$N = 2400$$

Expected values for CS412:

$$E_{412,A} = \frac{1370 \times 584}{2400} = 333.3$$

$$E_{412,B} = \frac{1370 \times 904}{2400} = 515.9$$

$$E_{412,C} = \frac{1370 \times 644}{2400} = 367.6$$

$$E_{412,D} = \frac{1370 \times 211}{2400} = 120.4$$

$$E_{412,F} = \frac{1370 \times 57}{2400} = 32.5$$

Expected values for CS512:

$$E_{512,A} = \frac{1030 \times 584}{2400} = 250.7$$

$$E_{512,B} = \frac{1030 \times 904}{2400} = 388.1$$

$$E_{512,C} = \frac{1030 \times 644}{2400} = 276.4$$

$$E_{512,D} = \frac{1030 \times 211}{2400} = 90.6$$

$$E_{512,F} = \frac{1030 \times 57}{2400} = 24.5$$

Now compute each chi-square term:

$$\chi^2 = \frac{(341 - 333.3)^2}{333.3} + \frac{(504 - 515.9)^2}{515.9} + \frac{(383 - 367.6)^2}{367.6} + \frac{(117 - 120.4)^2}{120.4} + \frac{(25 - 32.5)^2}{32.5}$$

$$+ \frac{(243 - 250.7)^2}{250.7} + \frac{(400 - 388.1)^2}{388.1} + \frac{(261 - 276.4)^2}{276.4} + \frac{(94 - 90.6)^2}{90.6} + \frac{(32 - 24.5)^2}{24.5}$$

$$\chi^2 \approx 9.98 \approx 10$$

Problem 2. Principle Component Analysis (PCA) (20 Points)

a) What is the data after standardization (Use the Population Level Equation)

$$X = \begin{bmatrix} 341 & 504 & 383 & 117 & 25 \\ 243 & 400 & 261 & 94 & 32 \end{bmatrix}$$

Population mean for each grade:

$$\mu_A = \frac{341 + 243}{2} = 292$$

$$\mu_B = \frac{504 + 400}{2} = 452$$

$$\mu_C = \frac{383 + 261}{2} = 322$$

$$\mu_D = \frac{117 + 94}{2} = 105.5$$

$$\mu_F = \frac{25 + 32}{2} = 28.5$$

Population standard deviation formula:

$$\sigma_j = \sqrt{\frac{(x_{1j} - \mu_j)^2 + (x_{2j} - \mu_j)^2}{2}}$$

Since there are only two values per column, each deviation is equal in magnitude:

$$\sigma_A = 49, \quad \sigma_B = 52, \quad \sigma_C = 61, \quad \sigma_D = 11.5, \quad \sigma_F = 3.5$$

Standardization formula:

$$z = \frac{x - \mu}{\sigma}$$

Standardized dataset:

$$Z = \begin{bmatrix} \frac{341-292}{49} & \frac{504-452}{52} & \frac{383-322}{61} & \frac{117-105.5}{11.5} & \frac{25-28.5}{3.5} \\ \frac{243-292}{49} & \frac{400-452}{52} & \frac{261-322}{61} & \frac{94-105.5}{11.5} & \frac{32-28.5}{3.5} \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix}$$

b) What is the covariance matrix of the standardized Dataset? (Use the Population Level Equation)

$$\Sigma = \frac{1}{N} Z^T Z$$

Since $N = 2$:

$$\Sigma = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & -1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}^T \begin{bmatrix} 1 & 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix}$$

c) What are the first principle component and second principal component of this dataset?

$$PC_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Any vector orthogonal to PC_1 (since rank = 1)

Problem 3. Correlation Analysis

Experiment	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Substance A	1.45	10.21	6.34	0.21	0.94	5.09	9.23	3.33	8.54	7.17
Substance B	61.3	45.2	98.9	23.4	50.6	87.4	5.3	78.4	23.2	85.3
Water Pollution	13.3	15.3	53.5	8.6	98.3	9	24.7	71.4	21.2	77.7

Experiment	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Substance A	1.11	4.15	3.85	54.8	0.12	4.82	9.65	8.21	3.12	6.23
Substance B	58.2	70.3	97.7	52.3	13.4	13.4	79.8	45.3	63.2	14.2
Water Pollution	13.4	5.1	15.7	23.5	81.4	37.2	73.6	83.6	45.7	9.3

Experiment	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
Substance A	8.93	2.82	6.23	7.86	7.34	1.53	8.2	6.74	4.23	8.24
Substance B	62.2	4.7	45.7	42.5	84.6	39.1	13.6	13.6	32.4	82.5

Experiment	21	22	23	24	25	26	27	28	29	30
Water Pollution	16	72.4	9.1	54.7	13.3	6.3	13.6	5.2	21	34.9

Experiment	31	32	33	34	35	36	37	38	39	40
Substance A	6.24	6.23	8.34	3.63	7.34	0.01	4.52	8.1	6.23	2.65
Substance B	51.5	24.6	84.2	37.6	64.1	18.5	2.4	26.7	44.2	34.1
Water Pollution	91.3	19.2	87.1	17.9	19.2	74	6.4	14.5	88.3	62.1

a) Separate the 40 experiments into two groups via a binning method where one group has low water quality (Group 1) and one group has high water quality (Group 2). Our goal here is to compare the correlation of substances A and B in these two groups to decide how to use substances A and B together to decrease water pollution. To do so, which binning method do you think is more appropriate here? Equal-width or equal depth? Why? Apply it to the dataset and create two groups.

- Equal-Width: Divide the range into bins of same size.
  - Problem: skewed data may give uneven group sizes.
- Equal-Depth : Divide so each bin has same number of experiments.
  - Advantage: groups are balanced.

I decided that the Equal-Depth binning method was more appropriate because it ensures that each group has roughly the same number of experiments

Experiment	Water Pollution
12	5.1
28	5.2
26	6.3
4	8.6
6	9.0
23	9.1
10	9.3
20	9.3
1	13.3
25	13.3
11	13.4
37	13.6
27	13.6
38	14.5
2	15.3
21	16.0
34	17.9
32	19.2
35	19.2
14	23.5
7	24.7
30	34.9
16	37.2
19	45.7
3	53.5
24	54.7
40	62.1
8	71.4
22	72.4
17	73.6
36	74.0
15	81.4
18	83.6
33	87.1
39	88.3
31	91.3
5	98.3

$$\text{Group 1 (Low water pollution)} : 5.1 - 24.7, \quad 20 \text{ experiments}$$

$$\text{Group 2 (High water pollution)} : 34.9 - 98.3, \quad 20 \text{ experiments}$$

b) Normalize the Substance A and Substance B amounts based on the min-max normalization (new range (0,1)) for each group and report the normalized values.

$$x_i^{\text{norm}} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$$

For Experiment 12, Substance A in Group 1:

- $x_i = 1.11$, $x_{\min} = 0.00$, $x_{\max} = 2.82$

$$A_{12}^{\text{norm}} = \frac{1.11 - 0.00}{2.82 - 0.00} \approx 0.394$$

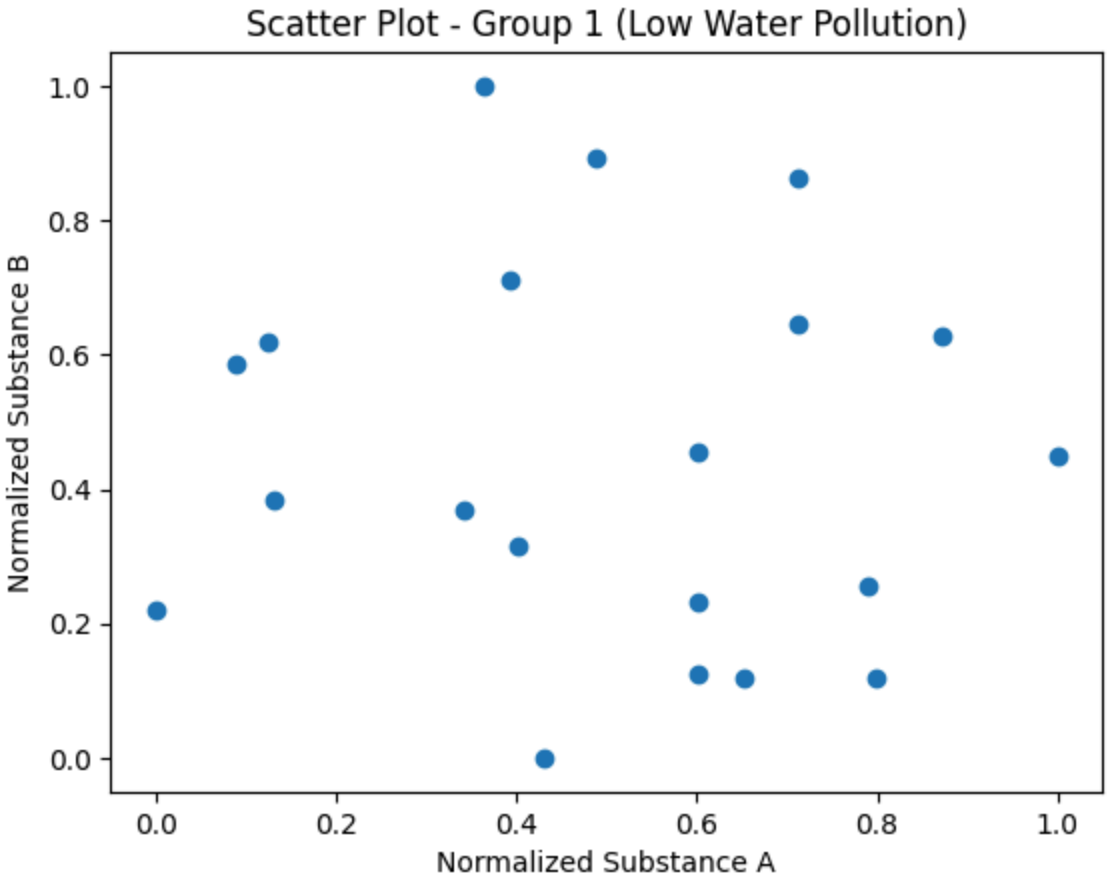
{Group 1 (Low Water Pollution) – Normalized Values

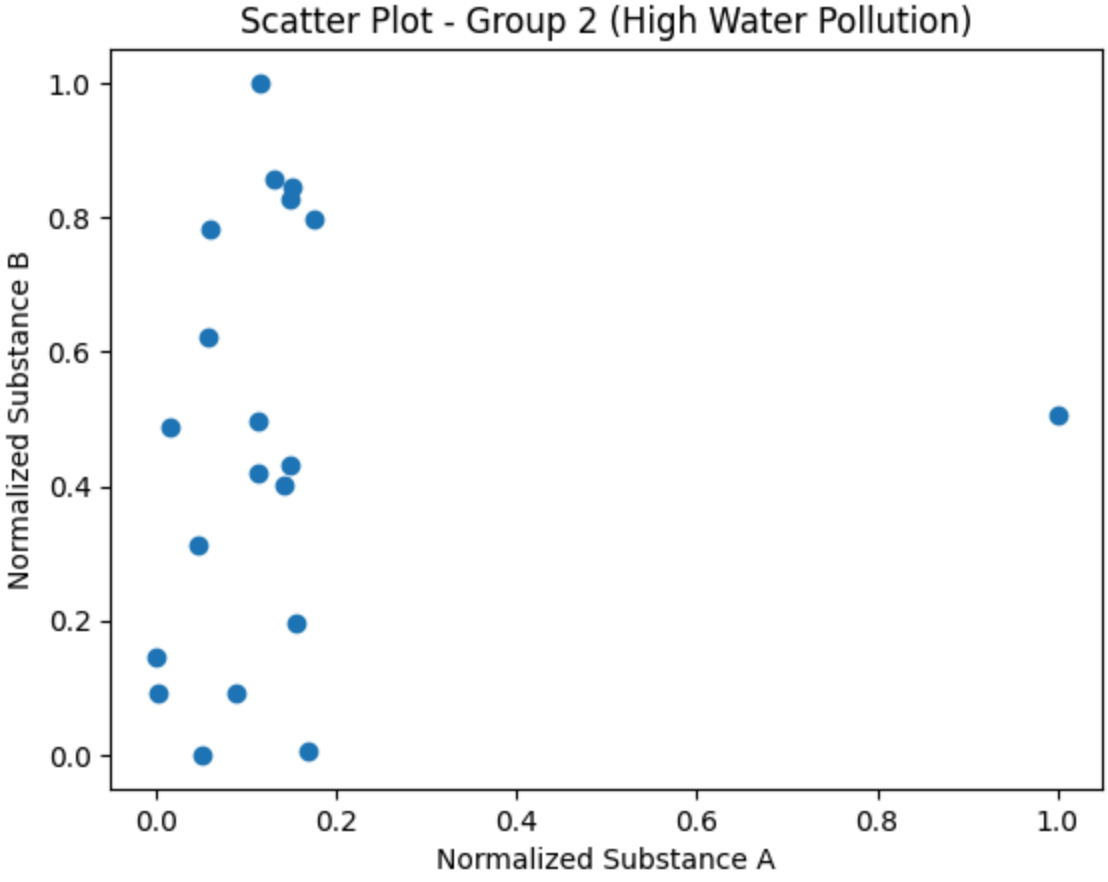
Experiment	A^{norm}	B^{norm}
12	$\frac{4.15-0.21}{10.21-0.21} = 0.394$	$\frac{70.3-2.4}{97.7-2.4} = 0.712$
28	$\frac{6.74-0.21}{10.21-0.21} = 0.653$	$\frac{13.6-2.4}{97.7-2.4} = 0.118$
26	$\frac{1.53-0.21}{10.21-0.21} = 0.132$	$\frac{39.1-2.4}{97.7-2.4} = 0.385$
37	$\frac{4.52-0.21}{10.21-0.21} = 0.431$	$\frac{2.4-2.4}{97.7-2.4} = 0.000$
4	$\frac{0.21-0.21}{10.21-0.21} = 0.000$	$\frac{23.4-2.4}{97.7-2.4} = 0.220$
6	$\frac{5.09-0.21}{10.21-0.21} = 0.488$	$\frac{87.4-2.4}{97.7-2.4} = 0.892$
23	$\frac{6.23-0.21}{10.21-0.21} = 0.602$	$\frac{45.7-2.4}{97.7-2.4} = 0.454$
20	$\frac{6.23-0.21}{10.21-0.21} = 0.602$	$\frac{14.2-2.4}{97.7-2.4} = 0.124$
25	$\frac{7.34-0.21}{10.21-0.21} = 0.713$	$\frac{84.6-2.4}{97.7-2.4} = 0.863$
1	$\frac{1.45-0.21}{10.21-0.21} = 0.124$	$\frac{61.3-2.4}{97.7-2.4} = 0.618$
11	$\frac{1.11-0.21}{10.21-0.21} = 0.090$	$\frac{58.2-2.4}{97.7-2.4} = 0.586$
27	$\frac{8.2-0.21}{10.21-0.21} = 0.799$	$\frac{13.6-2.4}{97.7-2.4} = 0.118$
38	$\frac{8.1-0.21}{10.21-0.21} = 0.789$	$\frac{26.7-2.4}{97.7-2.4} = 0.255$
2	$\frac{10.21-0.21}{10.21-0.21} = 1.000$	$\frac{45.2-2.4}{97.7-2.4} = 0.449$
13	$\frac{3.85-0.21}{10.21-0.21} = 0.364$	$\frac{97.7-2.4}{97.7-2.4} = 1.000$
21	$\frac{8.93-0.21}{10.21-0.21} = 0.872$	$\frac{62.2-2.4}{97.7-2.4} = 0.627$
34	$\frac{3.63-0.21}{10.21-0.21} = 0.342$	$\frac{37.6-2.4}{97.7-2.4} = 0.369$
35	$\frac{7.34-0.21}{10.21-0.21} = 0.713$	$\frac{64.1-2.4}{97.7-2.4} = 0.647$
32	$\frac{6.23-0.21}{10.21-0.21} = 0.602$	$\frac{24.6-2.4}{97.7-2.4} = 0.233$
29	$\frac{4.23-0.21}{10.21-0.21} = 0.402$	$\frac{32.4-2.4}{97.7-2.4} = 0.315$

Group 2 (High Water Pollutio) – Normalized Values

Experiment	A^{norm}	B^{norm}
9	$\frac{8.54-0.01}{54.8-0.01} = 0.156$	$\frac{23.2-4.7}{98.9-4.7} = 0.196$
14	$\frac{54.8-0.01}{54.8-0.01} = 1.000$	$\frac{52.3-4.7}{98.9-4.7} = 0.505$
7	$\frac{9.23-0.01}{54.8-0.01} = 0.168$	$\frac{5.3-4.7}{98.9-4.7} = 0.006$
30	$\frac{8.24-0.01}{54.8-0.01} = 0.150$	$\frac{82.5-4.7}{98.9-4.7} = 0.826$
16	$\frac{4.82-0.01}{54.8-0.01} = 0.088$	$\frac{13.4-4.7}{98.9-4.7} = 0.092$
19	$\frac{3.12-0.01}{54.8-0.01} = 0.057$	$\frac{63.2-4.7}{98.9-4.7} = 0.621$
3	$\frac{6.34-0.01}{54.8-0.01} = 0.116$	$\frac{98.9-4.7}{98.9-4.7} = 1.000$
24	$\frac{7.86-0.01}{54.8-0.01} = 0.143$	$\frac{42.5-4.7}{98.9-4.7} = 0.401$
40	$\frac{2.65-0.01}{54.8-0.01} = 0.048$	$\frac{34.1-4.7}{98.9-4.7} = 0.312$
8	$\frac{3.33-0.01}{54.8-0.01} = 0.061$	$\frac{78.4-4.7}{98.9-4.7} = 0.782$
22	$\frac{2.82-0.01}{54.8-0.01} = 0.051$	$\frac{4.7-4.7}{98.9-4.7} = 0.000$
17	$\frac{9.65-0.01}{54.8-0.01} = 0.176$	$\frac{79.8-4.7}{98.9-4.7} = 0.797$
36	$\frac{0.01-0.01}{54.8-0.01} = 0.000$	$\frac{18.5-4.7}{98.9-4.7} = 0.146$
10	$\frac{7.17-0.01}{54.8-0.01} = 0.131$	$\frac{85.3-4.7}{98.9-4.7} = 0.856$
15	$\frac{0.12-0.01}{54.8-0.01} = 0.002$	$\frac{13.4-4.7}{98.9-4.7} = 0.092$
18	$\frac{8.21-0.01}{54.8-0.01} = 0.150$	$\frac{45.3-4.7}{98.9-4.7} = 0.431$
33	$\frac{8.34-0.01}{54.8-0.01} = 0.152$	$\frac{84.2-4.7}{98.9-4.7} = 0.844$
39	$\frac{6.23-0.01}{54.8-0.01} = 0.114$	$\frac{44.2-4.7}{98.9-4.7} = 0.419$
31	$\frac{6.24-0.01}{54.8-0.01} = 0.114$	$\frac{51.5-4.7}{98.9-4.7} = 0.497$
5	$\frac{0.94-0.01}{54.8-0.01} = 0.017$	$\frac{50.6-4.7}{98.9-4.7} = 0.487$

c) Draw scatter plots of the normalized substance A and normalized substance B for each group


Scatter Plot - Group 1 (Low Water Pollution)
Normalized Substance B
Normalized Substance A
0.0
0.2
0.4
0.6
0.8
1.0


Scatter Plot - Group 2 (High Water Pollution)
Normalized Substance B
Normalized Substance A
0.0
0.2
0.4
0.6
0.8
1.0

d) Calculate the correlation coefficient (population level) between the normalized substance A and normalized substance B for each group.

Group 2 (High Water Pollution)

$$\mu_A = \frac{1}{20}(0.394 + 0.653 + 0.132 + 0.431 + 0.000 + 0.488 + 0.602 + 0.602 + 0.713 + 0.124 + 0.090 + 0.79$$

$$\mu_A = \frac{10.112}{20} = 0.506$$

$$\mu_B = \frac{1}{20}(0.712 + 0.118 + 0.385 + 0.000 + 0.220 + 0.892 + 0.454 + 0.124 + 0.863 + 0.618 + 0.586 + 0.1$$

$$\mu_B = \frac{8.985}{20} = 0.449$$

$$\rho = \frac{\sum_{i=1}^{20}(A_i - 0.506)(B_i - 0.449)}{\sqrt{\sum_{i=1}^{20}(A_i - 0.506)^2}\sqrt{\sum_{i=1}^{20}(B_i - 0.449)^2}}$$

$$\rho_{Group1} \approx 0.06$$

Group 1 (Low Water Pollution)

$$\mu_A = \frac{1}{20}(0.156 + 1.000 + 0.168 + 0.150 + 0.088 + 0.057 + 0.116 + 0.143 + 0.048 + 0.061 + 0.051 + 0.1$$

$$\mu_A = \frac{2.894}{20} = 0.145$$

$$\mu_B = \frac{1}{20}(0.196 + 0.505 + 0.006 + 0.826 + 0.092 + 0.621 + 1.000 + 0.401 + 0.312 + 0.782 + 0.000 + 0.79$$

$$\mu_B = \frac{8.310}{20} = 0.416$$

$$\rho = \frac{\sum_{i=1}^{20}(A_i - 0.145)(B_i - 0.416)}{\sqrt{\sum_{i=1}^{20}(A_i - 0.145)^2}\sqrt{\sum_{i=1}^{20}(B_i - 0.416)^2}}$$

$$\rho_{Group2} \approx -0.32$$

e) Based on the analysis in the above steps, draw a conclusion of how to use the two substances to decrease water pollution.

In the low pollution group, the correlation is close to 0.
This means Substance A and Substance B are not strongly related.
They can be used together without affecting each other much.

In the high pollution group, the correlation is negative.
This means when Substance A increases, Substance B tends to decrease.
They may interfere with each other in highly polluted water.

- For low pollution: use both substances together.
- For high pollution: adjust carefully and do not maximize both at the same time.

Problem 4. Discover Association Rules

Transaction	Items
T_1	{A, B, C, E}
T_2	{B, C, D}
T_3	{A, B, C, D}
T_4	{A, B, D, E}
T_5	{A, C, D}
T_6	{B, C, E}
T_7	{A, B, C, D, E}
T_8	{A, B, C}
T_9	{A, B, E}
T_10	{B, D, E}

Table 3: Transaction Data

(a) (10 points) Apply the Apriori algorithm to the transaction dataset in Table 3. Use a minimum support count of 2. Show all intermediate candidate sets Ck and frequent itemsets Lk for each k. Finally, list all frequent itemsets along with their support counts.

$$C_1 = \begin{array}{c|c} \text{Item} & \text{Support} \\ \hline A & 7 \\ B & 8 \\ C & 7 \\ D & 6 \\ E & 6 \end{array}$$

$$L_1 = \{A, B, C, D, E\}$$

$$C_2 = \begin{array}{|c|c|}\hline \text{Itemset} & \text{Support} \\ \hline AB & 6 \\ AC & 5 \\ AD & 4 \\ AE & 4 \\ BC & 6 \\ BD & 5 \\ BE & 5 \\ CD & 4 \\ CE & 3 \\ DE & 3 \\ \hline \end{array}$$

$$L_2 = C_2$$

$$C_3 = \begin{array}{|c|c|}\hline \text{Itemset} & \text{Support} \\ \hline ABC & 4 \\ ABD & 3 \\ ABE & 3 \\ ACD & 3 \\ ACE & 2 \\ ADE & 2 \\ BCD & 3 \\ BCE & 3 \\ BDE & 2 \\ CDE & 1 \\ \hline \end{array}$$

$$L_3 = \{ABC, ABD, ABE, ACD, ACE, ADE, BCD, BCE, BDE\}$$

$$C_4 = \begin{array}{|c|c|}\hline \text{Itemset} & \text{Support} \\ \hline ABCD & 2 \\ ABCE & 2 \\ ABDE & 2 \\ ACDE & 1 \\ BCDE & 1 \\ \hline \end{array}$$

$$L_4 = \{ABCD, ABCE, ABDE\}$$

$$L_5 = \varnothing$$

$$\text{All Frequent Itemsets} = L_1 \cup L_2 \cup L_3 \cup L_4$$

(b) (10 points) Apply the FP-Growth algorithm to identify frequent itemsets from a transactiondataset in Table 3. Show a clear FP tree for the dataset using a minimum support count of 2. Include the header table and node links. Then, show the step-by-step construction of thec onditional FP trees for generating frequent patterns. Finally, list all frequent itemsets with their support counts.

FP-Growth Frequency Table

Item	Support
B	8
A	7
C	7
D	6
E	6

Order: $B > A > C > D > E$

Final Frequent Itemsets (FP-Growth) $= L_1 \cup L_2 \cup L_3 \cup L_4$

